

# A Public Goods Property Mechanism for Decentralised Validator Networks

Research Proposal & Mechanism Design

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## Abstract

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We propose a novel public goods property mechanism for decentralised validator networks, combining NFT-based property rights, Harberger taxation, DAO-managed leasing, and guaranteed buy-back provisions. The mechanism is designed to eliminate speculative rent extraction from validator participation while preserving genuine price discovery, rewarding sustained service, and maintaining DAO solvency across variable network conditions. We situate the proposal within the intellectual tradition of Georgist land value theory, Radical Markets partial common ownership, and established proof-of-stake security models. We identify five core sub-problems — the speculation-participation dilemma, floor price solvency, supply equilibrium, tenure-seniority design, and slashing calibration — and set out the open mathematical questions whose resolution constitutes the research programme. The immediate application is the InfiniteZero Network (a trustless, privacy-first decentralised AI commons built on Ethereum), but the mechanism has broader implications for any shared productive infrastructure requiring enforceable, anti-speculative property norms.

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## I. The Core Problem Being Solved

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**Most blockchain validator systems generate an irresolvable tension between three properties that have never been simultaneously achieved: security, access, and accountability.**

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- **Security:** validators must have real skin-in-the-game to make honest participation the dominant strategy.
- **Access:** participation must not be gatekept by speculative asset prices that reflect scarcity rents rather than service value.
- **Accountability:** validators who misbehave or abandon their service obligations must bear real, proportionate consequences.

Token-based staking addresses security but generates rampant speculation, which in turn prices out genuine service providers and misaligns incentives toward asset appreciation rather than network contribution. The fundamental economic insight this mechanism imports from political economy — particularly from Georgist land theory and the RadicalXChange tradition — is that the right to participate in a commons is not the same as the right to speculate on its scarcity.

What we propose is a hybrid property regime: NFTs as licenced tenures over network roles, governed by the DAO in the manner a state governs land, with the Harberger tax as the anti-hoarding mechanism and a guaranteed buy-back as the anti-speculation floor. The mechanism is intended to ensure that those who contribute to the network earn reliably, can exit cleanly at the end of their service, and cannot profit merely from holding a scarce network position.

## II. Mechanism Overview

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The proposed mechanism consists of six interlocking components, each of which is described in turn. Taken together they constitute a programmable property regime for validator participation.

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### 1. NFT as Staked Property

Each validator NFT confers the right to occupy a productive role in the InfiniteZero Network, analogous to a property right in real estate. The NFT must be staked to activate participation. Misbehaviour results in slashing of the NFT's value; malicious conduct (two or three verified strikes) results in confiscation and return of the NFT to the DAO. The stake is not merely collateral — it is the property right itself, and its loss carries the same significance as a compulsory purchase or forfeiture in land law.

### 2. DAO Leasing and Lease Terms

The DAO issues validator NFTs as leases, much as a state government leases land for productive use. The initial lease term is ten years. At the end of the lease, the DAO exercises a mandatory buy-back of the NFT at its prevailing floor price, funded from accumulated network fees. The NFT may be transferred or sold at any point during the lease at or above its current floor price. The DAO controls total supply, issuing new NFTs or retiring expired ones according to observed network demand.

### 3. The Floor Price Engine

Each NFT carries a floor price that is set at the point of initial issuance and updated annually thereafter. The floor price tracks two variables: the validator's tenure (longer-staked validators earn a higher floor, reflecting demonstrated commitment and reduced counterparty risk) and network health (the floor is indexed to DAO fee revenue so that buy-back obligations remain solvent). The floor price is a guaranteed minimum exit value — it is the state's promise to the leaseholder that honest service will yield a recoverable capital value.

### 4. Harberger Taxation

Above the floor, the NFT's market price is subject to a Harberger tax on its self-assessed value. The holder declares a valuation; the tax rate  $\tau$  is applied to that valuation continuously; and any third party may purchase the NFT at the declared price at any

time. This creates the classical Harberger trade-off: over-valuation is penalised by higher taxes, and under-valuation is penalised by the risk of forced sale. The tax is paid from the validator's staking fee earnings, not from external capital, which ties the tax liability directly to the productive use of the network position.

## 5. Network Fee Distribution and Seniority

Validators earn a share of network fees in proportion to their staking activity and tenure. Seniority functions as a weighting factor: validators who have staked for longer are more likely to be selected for fee-generating services and receive a higher allocation from the fee pool. The DAO retains the remainder of fees to fund buy-backs and governance operations. This structure rewards loyalty and accumulated reputation without creating an unbounded incumbency advantage.

## 6. Partial Common Ownership and Auction Mechanics

In the event of a mandatory sale (triggered by a Harberger tax under-assessment), the NFT is transferred through an open auction to the highest bidder. This preserves allocative efficiency by ensuring network roles flow to those who value them most highly for productive use. Partial common ownership — where the DAO retains a residual interest through the lease structure — means that even during speculative episodes, the commons retains a claim on the underlying value of the role.

# III. The Five Core Sub-Problems

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The mechanism raises five interrelated sub-problems, each of which constitutes an open research question. We state them precisely here as the foundation for the mathematical programme in Section V.

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## Sub-Problem 1: The Speculation-Participation Dilemma

If validator NFTs are freely tradable and scarce, their market price will exceed the discounted value of expected service earnings, because purchasers will price in the option value of future scarcity. This crowds out validators who intend to earn by doing the work. The Harberger tax addresses this by making hoarding costly, but the optimal tax rate  $\tau^*$  is not known without empirical data on the network's production function.

Too low a rate permits persistent speculation. Too high a rate makes tenure so insecure that validators under-invest in quality, since they cannot expect to capture the returns on long-run improvements. This is the classical Harberger dilemma — allocative efficiency versus investment efficiency — and it has no closed-form resolution without specifying how validator quality maps to network output.

## Sub-Problem 2: Floor Price Solvency

The DAO's buy-back guarantee is a public put option. Its solvency requires that the net present value of fees collected over the lease period exceed the total buy-back liability at maturity. The floor price must therefore be a function of expected network revenue, not a fixed number set at genesis. This is the central actuarial problem of the mechanism: the DAO is issuing a future payment whose value depends on a revenue stream that is itself endogenous to participation.

Formally, for solvency we require:

$$\sum_t F(t) \cdot [\text{NFTs maturing at } t] \leq \sum_s R(s) / (1+\delta)^{(s-t)} \quad \text{for all future periods } t$$

where  $F(t)$  is the floor price at year  $t$ ,  $R(s)$  is DAO fee revenue at time  $s$ , and  $\delta$  is the discount rate. This condition must be maintained as a live on-chain invariant, not merely estimated at protocol genesis.

### Sub-Problem 3: Supply and Demand Equilibrium for Validator Slots

The number of active validator NFTs should reflect genuine service demand. Too few creates a bottleneck and a rent; too many dilutes per-validator fees to the point of non-participation. Unlike a conventional market where price signals supply, this mechanism fixes the price floor and controls supply through DAO governance. This creates a quantity-setting problem rather than a price-clearing problem: the DAO must determine issuance and retirement rules, ideally as a formalised algorithmic rule rather than continuous deliberation, which introduces governance latency and political risk.

### Sub-Problem 4: The Tenure-Seniority Incentive Structure

The seniority premium rewards demonstrated commitment but creates path-dependence: early entrants accumulate advantages that may become barriers to entry for later, potentially higher-quality validators. The seniority premium must be bounded to avoid calcifying into an incumbent oligopoly. The mathematical question is how to design a seniority function that asymptotes rather than compounds without limit — preserving legitimate tenure expectations without creating vested monopoly.

### Sub-Problem 5: Slashing Calibration

Slashing is the enforcement backbone of the system. The two- or three-strike rule introduces questions of severity weighting, decay over time, and the destination of slashed value (DAO treasury versus affected parties). Each choice changes the equilibrium behaviour of rational actors under the shadow of punishment. Too lenient a regime and the threat is not credible; too severe and validators price in slashing risk by under-staking, withdrawing at first dispute, or colluding to avoid triggering the rule.

## IV. Intellectual Lineage and Key Antecedents

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**The proposed mechanism sits at the intersection of four distinct traditions, each of which resolves part of the problem and creates a new one.**

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### Georgist Land Value Taxation

Henry George's foundational argument — that land rent belongs to the commons because it arises from collective action rather than individual effort — is the philosophical root of this proposal. The floor price and lease structure implement this directly: the DAO acts as the sovereign collecting ground rent, with validators as tenants paying for the right to occupy a productive role in the network.

### Harberger Taxation and Radical Markets

Glen Weyl and Eric Posner's Radical Markets formalised how self-assessment under mandatory-sale rules creates an equilibrium between the allocative efficiency of markets and the public extraction of monopoly rents. The canonical result is that with a tax rate  $\tau$ , the socially optimal allocation is achieved when  $\tau = 1$  (full socialisation), but investment efficiency requires  $\tau < 1$ . Empirical estimates from land markets suggest  $\tau$  in the range of 5–10% may be near-optimal, but the network context differs materially from land tenure and requires independent calibration.

### Proof-of-Stake Security Models

The standard PoS literature — from Buterin's slashing conditions to the Casper FFG model — provides the security-theoretic grounding for why stake must be at risk. The novel contribution here is replacing fungible token stake with an NFT whose value is itself endogenous to network health. This is a richer instrument than a fixed-quantity stake but requires the additional solvency analysis described above.

### Real-World Analogues: Singapore, Taiwan, and Norway

Singapore's 99-year land leaseholds demonstrate that state-managed buy-backs can coexist with a functioning price-discovery market. Norway's petroleum licensing rounds show how a sovereign can extract rent from a productive commons while sustaining private investment. Taiwan's quadratic financing experiments (supported by RadicalXChange and Vitalik Buterin) demonstrate the practical governance feasibility of partial common ownership mechanisms. The blockchain context compresses these timescales from decades to years and makes buy-back obligations programmable rather than political — both a strength (trustlessness) and a constraint (the DAO treasury must actually hold the requisite funds).

## V. The Open Mathematical Questions

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**We identify five open mathematical questions, ranked by importance for a first-principles treatment of the mechanism. Resolving these constitutes the technical core of the research programme.**

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## (a) Optimal Harberger Rate $\tau^*$

What tax rate maximises total social surplus — the sum of DAO revenue, validator earnings, and network user benefits — given the specific production function of the InfiniteZero Network? This requires specifying how validator quality maps to network output, which is an empirical question but can be parametrised for simulation and studied analytically in the neighbourhood of plausible calibrations.

$\max_{\tau} [ W_{\{DAO\}}(\tau) + W_{\{validator\}}(\tau) + W_{\{user\}}(\tau) ]$     subject to solvency constraint

## (b) Floor Price Update Rule $F(t)$

How should the floor price evolve with tenure, cumulative fee earnings, and network-level revenue? A candidate functional form is:

$$F(t) = F_0 \cdot (1 + \alpha \cdot \text{tenure}) \cdot (R(t) / R_0)^{\beta}$$

where  $\alpha$  controls the seniority premium and  $\beta$  controls network sensitivity. The parameters  $\alpha$ ,  $\beta$  must be chosen so that the DAO remains solvent under a range of network growth scenarios, including contraction. Stress-testing against adverse paths (fee decline, mass simultaneous lease expiry, tax revenue shock) is a prerequisite for deployment.

## (c) Supply Issuance Rule $N^*(t)$

Under what rule should the DAO issue or retire NFTs? A natural candidate is a target per-validator fee yield  $y^*$ : issue when yield exceeds  $y^* + \epsilon$ , retire when it falls below  $y^* - \epsilon$ , with a damping mechanism to prevent oscillation. The stability of this rule under rational expectations — validators will anticipate supply changes and adjust behaviour accordingly — must be formally verified.

$N^*(t)$ : issue if  $y(t) > y^* + \epsilon$  ; retire if  $y(t) < y^* - \epsilon$

## (d) Seniority Function $S(\text{tenure})$

How should the seniority premium scale with tenure? A logarithmic function bounds the premium and prevents oligopoly formation:

$$S(\tau) = 1 + \gamma \cdot \ln(1 + \tau)$$

where  $\gamma$  controls the degree of incumbency advantage permitted. The parameter  $\gamma$  must be calibrated so that seniority creates meaningful loyalty incentives without pricing out higher-quality entrants who arrive later in the network's lifecycle.

## (e) DAO Solvency Under Stress Scenarios

What is the probability of DAO insolvency — defined as inability to honour buy-back obligations — under adverse scenarios? The DAO's treasury balance is the state variable of a stochastic control problem, and we seek to characterise the steady state  $S^*$  where inflows (fees + tax) equal outflows (buy-backs + governance costs):

$$dS/dt = f(S, \text{participation rate}, \text{network activity}, \tau, F(t))$$

Steady state:  $S^*$  such that  $dS/dt = 0$  with  $P(\text{insolvency}) \leq \epsilon$

## VI. General Economic Contribution

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The most intellectually significant aspect of this mechanism is that it proposes a computable, enforceable property regime for shared productive infrastructure — a class of goods that conventional economics has handled poorly. Physical land is partially addressed by Georgism but is not programmable. Digital assets are programmable but have lacked enforceable property norms. The combination of smart-contract enforceability with the Harberger tax, seniority-weighted floors, and DAO buy-backs creates something without a precise real-world analogue:

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A public goods market with endogenous price floors, guaranteed exit liquidity, and anti-speculation enforcement.

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The closest conceptual relatives are affordable housing community land trusts (which use deed restrictions to maintain access without eliminating price signals) and spectrum licensing (which uses auctions and use-it-or-lose-it rules to prevent hoarding). Neither achieves all three properties simultaneously. This mechanism does so by making the guarantees programmable and the enforcement automatic, removing the political risk that undermines both analogue institutions in practice.

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The implications extend beyond blockchain. Any shared productive infrastructure — spectrum allocation, data licensing, municipal broadband, orbital slots, AI compute access — faces the same trilemma. The mathematical programme proposed here provides a formal framework for designing mechanisms that resolve it.

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## VII. Open Questions and Acknowledged Gaps

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We identify the following as genuine open problems requiring resolution before any mainnet deployment. These are not oversights but honest acknowledgements of the limits of the current design.

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- **Initial floor price setting:** The floor price cannot be determined by market discovery alone because there is no market before genesis. A combination of cost-of-service estimation, DAO governance, and an auction with a reserve price is likely required, with the auction as the primary mechanism and governance as a backstop.
- **Supply adjustment latency:** Any issuance rule requires DAO governance approval, which introduces delay. Rational validators will anticipate issuance decisions and front-run them, potentially destabilising the fee yield target. An automatic algorithmic rule is preferable but requires constitutional-level DAO commitment to prevent ex post override.
- **Network contraction risk:** The relationship between network health and buy-back obligations is the single most dangerous financial risk in the system. If fee revenue declines while large cohorts of leases approach maturity simultaneously, the DAO faces a solvency crisis. Dedicated actuarial modelling and a reserve fund protocol are required before deployment.
- **Harberger tax rate empirics:** The optimal  $\tau^*$  depends on empirical data about the network's production function that will not be available at genesis. A conservative starting rate with a governance-controlled update mechanism — constrained by a constitutional upper bound — is the safest initial design.
- **Validator collusion:** Seniority weighting creates an incentive for incumbent validators to collude in excluding new entrants, particularly near supply capacity. This requires an explicit anti-collusion protocol layer, potentially including a randomised allocation element in fee distribution.

## VIII. Proposed Research Programme

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We propose the following research programme to formalise the mechanism and prepare it for deployment:

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### Phase 1: Formal Economic Specification (Months 1–3)

- Derive the optimal  $\tau^*$  analytically under simplified production function assumptions.
- Specify the floor price update rule  $F(t)$  and prove the solvency condition holds over plausible parameter ranges.
- Formalise the supply issuance rule  $N^*(t)$  and verify stability under rational expectations.
- Define the seniority function  $S(\tau)$  and bound the incumbency advantage.

### Phase 2: Agent-Based Simulation (Months 4–6)

- Build an agent-based model of validator behaviour under the mechanism, including rational actors, strategic players, and noise traders.
- Simulate supply and seniority dynamics across growth and contraction scenarios.
- Stress-test DAO solvency against adverse parameter paths.
- Calibrate  $\alpha$ ,  $\beta$ ,  $\gamma$ , and  $\tau$  against simulation outcomes.



### Phase 3: Whitepaper and Testnet Deployment (Months 7–12)

- Produce a complete technical whitepaper with formal proofs and simulation results.
- Deploy the mechanism on an Ethereum testnet, implementing the smart contracts in the existing DIN-Protocol-Toolkit.
- Conduct a structured user study with prospective validators to test behavioural assumptions.
- Submit findings to an academic venue (e.g. Financial Cryptography, EC, ACM CCS).

## IX. Conclusion

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**The mechanism described in this proposal is coherent, grounded in serious economic theory, and meaningfully novel in its combination of instruments. It addresses a real failure mode in existing validator systems — the speculation-participation dilemma — using tools from political economy that have been theoretically validated but never implemented in a fully programmable, trustless environment.**

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**The InfiniteZero Network provides the ideal deployment context: a genuine public goods mission (decentralised AI commons), an Ethereum-native architecture, and an existing community of validators and developers committed to the vision. The research programme proposed here — formal specification, agent-based simulation, and testnet deployment — represents the minimum viable path from conceptual design to rigorous, deployable protocol.**

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**We invite collaboration, critique, and funding support to carry this programme forward. Contact: [abrahamnash@protonmail.com](mailto:abrahamnash@protonmail.com) — InfiniteZero Foundation, Oxford.**

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